SNOWPRO® ADVANCED: ARCHITECT EXAM STUDY GUIDE

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SNOWPRO ADVANCED: ARCHITECT STUDY GUIDE OVERVIEW

This is a self-learning study guide that highlights concepts that may be covered on Snowflake's SnowPro Advanced: Architect Certification exam.

Using this study guide does not guarantee certification success.

Holding the SnowPro Core certification in good standing is a prerequisite for taking the Advanced: Architect certification.

For an overview and more information on the SnowPro Core Certification exam or SnowPro Advanced Certification series, please navigate here.

RECOMMENDATIONS FOR USING THE GUIDE

This guide will show the Snowflake topics and subtopics covered on the exam. Following the topics will be additional resources consisting of videos, documentation, blogs, and/or exercises to help you understand architecting on Snowflake.

Estimated length of time required to complete the study guide: 10 – 13 hours

SNOWPRO ADVANCED: ARCHITECT CERTIFICATION OVERVIEW

The **SnowPro Advanced: Architect** tests advanced knowledge and skills used to apply comprehensive architect solutions using Snowflake. The exam will assess skills through scenario-based questions and real-world examples.

This certification will test the ability to:

- Design an end-to-end data flow from source to consumption using the Snowflake Platform
- Design and deploy a data architecture that meets business, security, and compliance requirements.
- Select appropriate Snowflake and third-party tools to optimize architecture performance.
- Design and deploy a shared data set using the Snowflake Marketplace and Data Exchange.

Target Audience:
2+ years of practical experience with Snowflake as an Architect in a production environment. In addition, successful candidates may have:

- Hands-on expertise with SQL and SQL analytics
- Experience building out a complex ETL/ELT pipeline
- Experience implementing security and compliance requirements
- Working with different data modeling techniques.

Having coding experience outside of SQL and DevOps/DataOps design experience is a plus.

This exam is designed for:

- Solution Architects
- Database Architects
- System Architects

SNOWPRO ADVANCED: ARCHITECT PREREQUISITE

Eligible individuals must hold an active SnowPro Core Certified credential. If you feel you need more guidance on the fundamentals, please see the SnowPro Core Exam Study Guide.

STEPS TO SUCCESS

1. Review the Advanced: Architect Exam Guide
    - Get started with Snowflake Quick Starts
    - Review recommended Snowflake Documentation
2. For additional training, attend Snowflake's Instructor-Led Architect Course
3. Take the Snowflake Well Architected On-Demand Training
4. Review and study applicable white papers and documentation
5. Get hands-on practical experience with relevant business requirements using Snowflake
6. Attend Snowflake Webinars
7. Practice with sample questions here
8. Attend Virtual Hands-on Labs for more hands-on practical experience
9. Schedule your exam
10. Take your exam!

SNOWPRO ADVANCED: ARCHITECT SUBJECT AREA BREAKDOWN

This exam guide includes test domains, weightings, and objectives. It is not a comprehensive listing of all the content that will be presented on this examination. The table below lists the main content domains and their weightings.

Domain	Domain Weightings on Exams
1.0 Accounts and Security	25%
2.0 Snowflake Architecture	30%
3.0 Data Engineering	25%
4.0 Performance Optimization	20%

Domain 1.0: Account and Security

1.1 Design a Snowflake account and database strategy, based on business requirements.

- Create and configure Snowflake parameters based on a central account and any additional accounts.
  - Parameters (all levels)
    - Account parameters
    - Object parameters
    - Session parameters
  - Outline the Snowflake parameter hierarchy and the relationship between the parameter types.
- List the benefits and limitations of one Snowflake account as compared to multiple Snowflake accounts.
  - Isolate or segment accounts
  - Key considerations and constraints when defining an account strategy
  - Features/capabilities that can be leveraged across accounts
  - Identify use cases that are appropriate for account strategies

1.2 Design an architecture that meets data security, privacy, compliance, and governance requirements.

- Configure Role-Based Access Control (RBAC) hierarchy
  - Privilege inheritance
  - Database roles
  - System roles and associated best practices
  - Functional roles compared to access roles
  - Secondary roles
- Data Access
  - Storage integrations
- Data Security
  - Secure views
  - Data Governance
    - Column-level security
      - External tokenization
      - Dynamic Data Masking
    - Row-level security
      - Row access policies
    - Aggregate policies
    - Projection policies
    - Data lineage and dependencies
    - Object tagging
  - Compliance
  - Features of the different Snowflake editions
    - Payment Card Industry (PCI) Security Standard
    - Personal Identifiable Information (PII)/ Personal Health Information (PHI)

1.3 Outline Snowflake security principles and identify use cases where they should be applied.

- Encryption
- Network security
  - Network policies
  - Network rules
  - External access
  - Access control privileges
  - Private connectivity
    - AWS PrivateLink
    - Azure Private Link
    - Google Cloud Private Service Connect
- User, role, and grants provisioning
- Authentication
  - Authentication policies
  - Federated authentication
  - Single Sign-On (SSO)
  - OAuth
  - Multi-Factor Authentication (MFA)
  - Key-pair authentication
  - Security integration

Domain 1.0 Study Resources

Lab Guides

Architecture Pattern: Security – Access to Sensitive Objects
Architecture Pattern: Security – Authentication
Architecture Pattern: Security – Network

Additional Assets

Cloud Data Platform Security: How Snowflake Sets the Standard (white paper)
Snowflake for Data Mesh
Snowflake Pattern - Security - Access to Sensitive Objects
Snowflake Query Timeout: Resolving STATEMENT_TIMEOUT_IN_SECONDS Conflicts
Snowflake Security Overview and Best Practices

Snowflake Documentation

About Secure Data Sharing
Access control best practices
Access control privileges
Advanced Column-level Security topics
ALTER USER
AWS PrivateLink and Snowflake
Cloning considerations
Configuring an identity provider (IdP) for Snowflake
Controlling network traffic with network policies
CREATE <object> … CLONE
CREATE STORAGE INTEGRATION
DEVOPS FOR DATA APPS ON SNOWFLAKE
External network access examples
GET_DDL
How To: configure SAML SSO authentication with Okta
Introduction to external functions
Introduction to object tagging
Introduction to organizations
Key-pair authentication and rotation
Manage reader accounts
Managing/Using federated authentication
Network rules
Okta SCIM integration with Snowflake
Overview of Access Control
Overview of federated authentication and SSO
Parameters
Power BI SSO to Snowflake
Share data securely across regions and cloud platforms
Snowflake Editions
SYSTEM$CLASSIFY
SYSTEM$GLOBAL_ACCOUNT_SET_PARAMETER
Transactions
Understanding caller's rights and owner's rights stored procedures
Understanding Column-level Security
Understanding Encryption Key Management in Snowflake
Understanding External Tokenization
Understanding row access policies
USE SECONDARY ROLES
Using Dynamic Data Masking

Domain 2.0: Snowflake Architecture

2.1 Outline the benefits and limitations of various data models in a Snowflake environment.

- Data models
  - Data vault
  - Star schema
- Use of key/column constraints (ENABLE/RELY/VALIDATE)

2.2 Design data sharing solutions, based on different use cases.

- Use cases
  - Sharing within the same organization/same Snowflake account
  - Sharing within a cloud region
  - Sharing across cloud regions
  - Sharing between different Snowflake accounts
  - Sharing to a non-Snowflake customer
  - Sharing across cloud providers
  - Sharing using Snowflake Data Clean Rooms
- Snowflake Marketplace
- Data Exchange
- Data sharing methods
  - Configure shares, account parameters, and privileges
  - Security patterns for data sharing
  - Outline the purpose, benefits, and capabilities of the multiple data sharing methods
  - Cross-Cloud Auto-Fulfillment

2.3 Create architecture solutions that support development lifecycles as well as workload requirements.

- Data lake and environments
  - Storage directory structure
  - Zones (data warehouse layers)
  - Support of DevOps/DataOps principles
  - Production/development/sandbox
  - Data workloads
  - Data warehouse
  - ELT/ETL
- Development lifecycle support
  - Migration
  - Deployment
    - CI/CD
    - Snowflake CLI
  - Git integration
  - Rollback process
- Outline basic AI/ML pipelines and applications
  - Snowpark Container Services
  - Snowflake ML functions
  - Cortex LLM functions
  - Streamlit
  - Snowflake Native App Framework

2.4 Given a scenario, outline how objects exist within the Snowflake object hierarchy and how the hierarchy impacts an architecture.

- Roles
- Virtual warehouses
- Object hierarchy
  - Databases
    - Schemas
      - Tables
      - Views
      - Stages
- File formats
- Functions
- Procedures
- Streams and tasks

2.5 Determine the appropriate data recovery solution in Snowflake and how data can be restored.

- Backup/recovery
  - Time Travel
    - Table types
    - Costs
    - Availability
    - Query performance impacts
  - Data corruption impacts
  - Zero-copy cloning
  - Fail-safe
- Disaster recovery
  - Replication and failover

Domain 2.0 Study Resources

Lab Guides

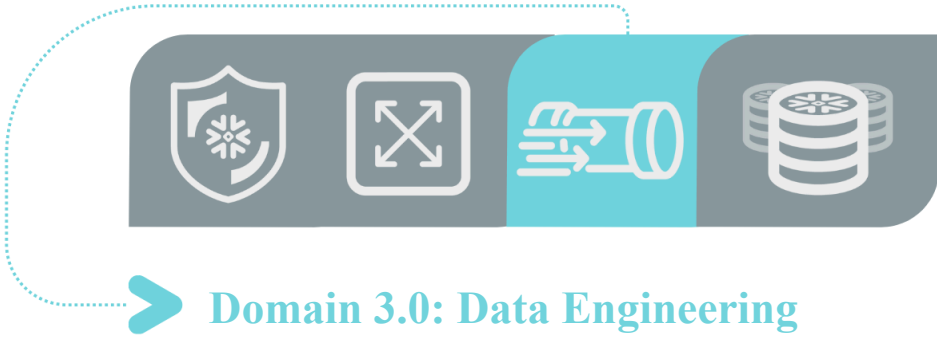
Building an application on Snowflake with data from Snowflake Data Marketplace
Getting Started with Time Travel
Snowflake and Data Modeling (workload guide)

Additional Assets

IOT Reference Architecture (use case)
Application Health and Security Analytics Reference Architecture (use case)
Multiple Data Modeling Approaches with Snowflake (blog)
Data Modeling in the Age of JSON and Schema-on-Read (blog)
Undrop object dropped or replaced multiple times (blog)
Support Multiple Data Modeling Approaches with Snowflake (blog)
How to restore a table with dropped column(s) using Time Travel with cloning (article)
Data Clean Room Live Demo: Improve Advertising Effectiveness and Data Security and Privacy Compliance (video)
Data Vault Techniques on Snowflake: Conditional Multi-Table INSERT, and Where to Use It (blog)
Iceberg Tables & Snowflake. The importance of table formats and… | by Daniel Myers | Snowflake Builders Blog: Data Engineers, App Developers, AI/ML, & Data Science (blog)
Modernizing Data Warehousing with Snowflake and Hybrid Data Vault (blog)
Snowflake for Data Mesh (use case)
Snowflake Pattern - Security - Access to Sensitive Objects (blog)
What is a Data Clean Room, and Do You Need One? (blog)

Snowflake Documentation

About Data Exchange
About listings
About Secure Data Sharing
ALTER SCHEMA
Auto-fulfillment for listings
Cloning considerations
CREATE <object> … CLONE
CREATE CONNECTION
CREATE DATABASE
CREATE DATABASE ROLE
Database replication considerations
Failing over account objects
GET_DDL
GRANT DATABASE ROLE
GRANT OWNERSHIP
Introduction to external functions
Introduction to external tables
Introduction to replication and failover across multiple accounts
Overview of Access Control
Parameters
Redirecting client connections
Replication considerations
Share data from multiple databases
Share data protected by a policy
Share data securely across regions and cloud platforms
Share secure database objects
Snowflake Editions
Transactions
Understanding & using Time Travel
Understanding caller's rights and owner's rights stored procedures
Understanding row access policies
UNDROP DATABASE
UNDROP TABLE
Working with Secure Views
Working with Temporary and Transient Tables



Domain 3.0: Data Engineering

3.1 Determine the appropriate data loading or data unloading solution to meet business needs.

- Data sources
  - Data at rest
  - Data in motion
  - External sources and formats
  - Streaming data
    - Snowpipe
    - Change Data Capture (CDC)
  - OLTP/RDBMS sources
  - API sources
- Data ingestion
  - Bulk file upload
  - Snowpipe
  - Snowpipe Streaming
  - External tables
  - Reload process (load history)
  - Incremental updates compared to full updates
  - Iceberg tables (managed and unmanaged)
  - Parameters for copying data and addressing data handling errors
- Architecture changes
  - Schema detection and table schema evolution
  - Data source changes
- Data unloading

3.2 Outline key tools in Snowflake's ecosystem and how they interact with Snowflake.

- Connectors
  - Kafka
  - Spark
  - Python
  - Snowflake Connector for ServiceNow
  - Snowflake Connector for Google Analytics
- Drivers
  - JDBC
  - ODBC
- API endpoints
  - Use of `system$allowlist`
  - SQL API
- SnowSQL
- Snowflake CLI
- Snowpark
  - Python
  - Scala
  - Java

3.3 Determine the appropriate data transformation solution to meet business needs.

- Views and tables
  - Benefits, limitations, properties
  - Relationship and impact between the view and data types
  - Impact of costs
  - Dynamic tables
- Staging layers and tables
- Querying semi-structured data
  - Flatten
- Data processing
- Stored procedures
- Streams and tasks
- Functions
  - External functions
    - Performance impacts
  - User-Defined Functions (UDFs)
  - User-Defined Table Functions (UDTFs)
  - Secure functions

Domain 3.0 Study Resources

Lab Guides

Building a Real-Time Data Vault in Snowflake
Enrich Salesforce data with Snowflake to deliver your Customer 360
Auto-Ingest Twitter Data into Snowflake
Build a Recommendation Engine with AWS SageMaker
Getting Started with Snowpipe
Accelerating Data Engineering with Snowflake & dbt
Getting Started With User-Defined Functions
Getting Started with Python
Getting Started with SnowSQL
Accelerating Data Teams with dbt Cloud & Snowflake

Additional Assets

Snowflake for Data Engineering- Easily Ingest, Transform, and Deliver Data for Up-To-The Moment Insights (white paper)
Masking Semi-Structured Data with Snowflake (blog)
Easy Continuous Data Pipelines with GA of Streams and Tasks (blog)
Best Practices for Data Unloading
How to restore a table with dropped column(s) using Time Travel with cloning (article)
Iceberg Tables & Snowflake (blog)
Data Vault Techniques on Snowflake: Conditional Multi-Table INSERT, and Where to Use It (blog)

Snowflake Documentation

Add a Streamlit app
ALTER TABLE
Automatic Clustering
Automating continuous data loading using cloud messaging
Automating Snowpipe for Amazon S3
Clustering Keys & Clustered Tables
Considerations for semi-structured data stored in VARIANT
COPY INTO <location>
COPY INTO <table>
Copying data from an S3 stage
CREATE DYNAMIC TABLE
CREATE ICEBERG TABLE (Iceberg files in object storage)
CREATE PIPE
Data unloading considerations
Event table overview
EXPLAIN
FLATTEN
GET
Introduction to SQL UDFs
Introduction to Streams and Tasks
Introduction to unstructured data
Managing Snowpipe
MERGE
Overview of data loading
Overview of the Kafka connector
Overview of warehouses
PUT
Querying Semi-structured Data
RESULT_SCAN
Search Optimization Service
SENTIMENT (SNOWFLAKE.CORTEX)
Setting up Snowflake to use Git
Snow git commands
Snowflake Connector for Python
Snowflake Cortex AISQL (including LLM functions)
Snowflake Python APIs: Managing Snowflake objects with Python
Snowpark Container Services
Snowpipe
Snowpipe Streaming
Transforming data during a load
User-defined functions overview
Using Snowflake Connector for Kafka with Snowpipe Streaming Classic
Using the Python Connector
Using the Query Acceleration Service
Views, materialized views, and dynamic tables
Warehouse considerations
Working with resource monitors

Domain 4.0: Performance Optimization

4.1 Outline performance tools, best practices, and appropriate scenarios where they should be applied.

- Query profiling
  - Interpret a Query Profile, identify bottlenecks, and outline recommendations
  - Metadata functions
  - Warehouse queuing
  - Warehouse spilling
- Virtual warehouse configurations
  - Auto-suspend/resume
  - Scale up/down (resizing)
  - Scale in/out (multi-cluster warehouse/auto-scaling)
  - Query acceleration service
  - Snowpark-optimized warehouses
- Clustering
  - Natural clustering
  - Auto-clustering
  - Clustering keys
- Search optimization service
- Caching
  - Different cache layers
  - Cache expiration
  - Impact of costs

4.2 Troubleshoot performance issues with existing architectures.

- Use of system clustering information
- Warehouse monitoring
- Optimization techniques
- Micro-partition pruning
- Monitoring and alerting
  - `ACCOUNT_USAGE` and `INFORMATION_SCHEMA` views
  - Resource monitoring
  - Alerts and notifications (for example, errors, email)
  - Event tables (for example, logging, tracing

Domain 4.0 Study Resources

Lab Guides

Resource Optimization: Performance
Resource Optimization: Billing Metrics
Resource Optimization: Setup & Configuration
Resource Optimization: Usage Monitoring
Automating Data Pipeline to Drive Marketing Analytics with Snowflake & Fivetran
Building a Data Application
DevOps: Database Change Management with schemachange and GitHub

Additional Assets

Customer 360 Reference Architecture (use case)
The Rise of Data Exchanges Frictionless Integration of Third-Party Data (white paper)
Definitive Guide to Managing Spend in Snowflake (white paper)

Snowflake Documentation

Collecting metrics data
CREATE TASK
Dynamic tables
HLL
Logging and tracing: Default event table included
Micro-partitions & Data Clustering
Monitoring search optimization using Snowsight
Multi-cluster warehouses
Optimizing the warehouse cache
Set up event tracing for an app
TABLES view
Transactions - Long-running Statements
Working with Materialized Views
Working with Subqueries

SNOWPRO ADVANCED: ARCHITECT SAMPLE QUESTIONS

1. An Architect is optimizing a query performance. The query is reading data from a clustered fact table, causing remote spillage and inadequate pruning.

   Which system function should be used to show the partition depth histogram?

   A. `SYSTEM$CLUSTERING_DEPTH`

   B. `SYSTEM$CLUSTERING_RATIO`

   C. `SYSTEM$CLUSTERING_INFORMATION`

   D. `SYSTEM$CLUSTERING_HISTOGRAM`

2. A user has a table named `CURRENT_EMPLOYEE` and would like to create a new table named `CURRENT_EMPLOYEE_COPY_DATA`, copying only the table structure from the `CURRENT_EMPLOYEE` table.

   Which query should be run?

   A. `CREATE TABLE current_employee_copy_data COPY current_employee;`

   B. `CREATE TABLE current_employee_copy_data LIKE current_employee;`

   C. `CREATE TABLE current_employee_copy_data CLONE current_employee;`

   D. `CREATE TABLE current_employee_copy_data as SELECT * from current_employee;`

3. Person1 is using the role `SECURITYADMIN`. Person 1 creates a role named `DBA_ROLE` that will be used to manage the warehouses in the Snowflake account. Person1 now needs to switch to that role.

   What command(s) must be executed in order to switch the context of this worksheet?

   A. `USE ROLE DBA_ROLE;`

   B. `GRANT ROLE DBA_ROLE TO ROLE SECURITYADMIN;`

   C. The `SECURITYADMIN` role is not allowed to `GRANT` permissions to a role.

   D. `GRANT ROLE DBA_ROLE TO USER PERSON1; USE ROLE DBA_ROLE;`

4. A table is created using the following syntax:

```
CREATE TABLE MY_TABLE (NAME STRING(100));
```

What would the command `DESC TABLE MY_TABLE;` display as the column type?

A. String

B. Char

C. Text

D. Varchar

5. How can data sharing be used across regions and cloud platforms? (Select TWO).

A. Data can be shared across regions using database replication, but not across different cloud providers.

B. Data can be replicated to a different region and a share can be created on that replica.

C. A share cannot be created on a replica, so sharing across regions and cloud platforms is not possible.

D. Data can be cloned across regions and a share can be created on the clone.

E. Data can be replicated to a different cloud provider and a share can be created on the replica.

Correct responses for sample questions:

1: c, 2: b, 3: d, 4: d , 5: b & e

NEXT STEPS

REGISTERING FOR YOUR EXAM

When you are ready to register for the exam navigate here to get started. Select the exam you want to take and click "Register Now". This will take you to our Certification Management system where you will register to take the exam.

MAINTAINING YOUR CERTIFICATION

All Snowflake Certifications expire two (2) years after your certification issue date.

SnowPro Certifications can now be recertified through the Snowflake Continuing Education (CE) program which includes these options -

- Completion of eligible Snowflake Instructor Led (ILT) Training Courses
- Earning of an equivalent or higher-level SnowPro Certification

Note: You must have a valid Certification to participate in the Continuing Education (CE) program.

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